Project Report

E-Commerce Return Risk Intelligence Dashboard

1. Introduction

In the fast-growing e-commerce industry, product returns present significant operational and financial challenges. High return rates increase logistics costs, reduce profitability, and negatively affect customer experience. Identifying the factors that drive returns and predicting high-risk orders in advance can help organizations take preventive and data-driven actions.

This project focuses on analyzing e-commerce return behavior and building a predictive return risk system. Using historical order and customer data, the project develops a return risk score and presents insights through an interactive Power BI dashboard to support business decision-making and return reduction strategies.

2. Abstract

The E-Commerce Return Risk Intelligence project analyzes customer return patterns using transactional e-commerce data. The project integrates data preprocessing, exploratory data analysis, predictive modeling, and business intelligence visualization.

A logistic regression model is used to estimate the probability of return for each order. High-risk orders are identified using a percentile-based threshold to address class imbalance. The results are visualized in an interactive Power BI dashboard featuring return risk KPIs, high-risk product tables, and drill-through analysis. This solution enables proactive return management and supports operational, marketing, and product-level decisions.

3. Tools Used

- **Python**
  - Pandas, NumPy for data cleaning and analysis
  - Scikit-learn for logistic regression modeling
- **Power BI Desktop**
  - Data transformation and modeling
  - DAX measures for KPIs
  - Interactive dashboards with slicers and drill-through
- **CSV Dataset (Kaggle E-commerce Data)**

    - Orders, products, customers, and return information

- **SQL (Conceptual Use)**
    - Designing analytical views for reporting

4. Steps Involved in Building the Project

Step 1: Data Understanding and Cleaning

The dataset was analyzed to understand order details, customer attributes, product categories, and return status. Data cleaning included handling missing values, converting date fields, removing duplicates, and creating a binary return indicator.

Step 2: Exploratory Data Analysis (EDA)

EDA was conducted to study return rates across categories, regions, and payment methods. This analysis helped identify key return drivers such as delivery delays and category-specific issues.

Step 3: Feature Engineering

New features such as delivery lead time and payment indicators were created to improve model performance and enhance business relevance.

Step 4: Predictive Modeling

A logistic regression model was trained to predict return probability for each order. A percentile-based approach was used to classify the top 10% of orders as high-risk.

Step 5: Dashboard Development

A Power BI dashboard was developed with return risk KPIs, high-risk products tables, category and region analysis, and drill-through functionality for product-level insights.

5. Conclusion

This project demonstrates the practical application of data analytics and predictive modeling to address a real-world e-commerce problem. By identifying return drivers and predicting high-risk orders, businesses can take proactive steps to reduce return-related losses.

The Return Risk Intelligence Dashboard combines machine learning outputs with intuitive visualizations, providing actionable insights for decision-makers. The project highlights strong skills in data analysis, basic machine learning, and business intelligence, making it suitable for academic evaluation, internships, and entry-level data analyst roles.